MIL-C-83500(USAF)
AMENDMENT 2
16 JUNE 1980
SUPERSEDING
AMENDMENT 1
3 FEB 1976

MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM CATHODE, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-83500(USAF), dated 19 June 1975.

PAGE 1

Title, delete and substitute: "CAPACITORS, FIXED, ELECTROLYTIC, (NONSOLID ELECTROLYTE), TANTALUM ANODE AND CATHODE, GENERAL SPECIFICATION FOR"

1.1, line 1, following "Tantalum", add "anode and."

PAGE 11

TABLE III, Subgroup 2, before "Life at 85°C (240-hour)" add:

"

Test	Requirement paragraph	Test paragraph
Thermal shock 1/	3.11	4.6.8

1/ Twenty five cycles.

"

PAGE 13

TABLE IV: Subgroup 5, delete "every 4 months" and substitute "every 12 months".

Subgroup 6, delete and substitute:

"

Test	Requirement paragraph	Test paragraph
Barometric pressure (reduced)	3.20	4.6.17
Life at 85°C	3.22.3	4.6.19.3

"

PAGE 14

4.6.7(a), delete and substitute:

"(a) Mounting of specimens - External mounting means."

PAGE 15

4.6.8(c), delete and substitute:

"(c) Number of cycles - 300 (except group B inspection - 25 cycles)."

PAGE 17

* TABLE V: Delete title and substitute "DC surge voltages, derated surge voltages, and rated voltages."

* TABLE V, Column 2: Delete "Derated voltage at 125°C" and substitute "Derated surge voltage at 125°C."

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* 4.6.15, delete and substitute:

"4.6.15 Seal (see 3.18). Capacitors shall be tested in accordance with method 112 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - C.

(b) Degree of leakage rate sensitivity -- 10^{-8} atm cc/sec.

(c) Procedure -- The mass-spectrometer-type leak detector shall be coupled to the unsealed evacuation port of the specimen, and a vacuum created within the specimen. It is extremely important that the coupling connections between the specimen and the leak detector be perfectly sealed. The specimen shall then be subjected to a gas atmosphere either by surrounding the specimen with the gas or by spraying the specimen thoroughly with a jet of the gas. If a defect exists in the specimen, an amount of gas that depends upon the size of the defect will be drawn through it and passed into the leak detector, which will read the leakage rate.

(d) Measurements after test -- Not applicable.

Following this test, capacitors shall be visually examined for evidence of leakage."

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* 4.6.20 (b), delete and substitute:

"(b) Operating conditions-derated voltage (see figure 2) shall be applied."

PAGE 21

* 4.6.22, delete and substitute:

"4.6.22 AC ripple life at 125°C (see 3.25). Capacitors shall be tested as specified in 4.6.19.1, with the following exceptions:

(a) Test temperature and tolerance -- 125°+4°,-0°C

(b) Operating conditions -- Capacitors shall be tested at 40 ±2 kHz sinusoidal and 750 mA rms (±10 percent). The peak ac voltage plus dc shall not exceed the rated voltage of the capacitor."

PAGE 22

* 6.3, delete and substitute:

"6.3. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force Acquisition Logistics Division, Electronic Support Division (AFALD/PTES), Wright-Patterson Air Force Base, Ohio 45433; however, informatin pertaining to qualification of products covered by this specification may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444."

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Add the following new paragraph:

"6.5 Mounting of units. Mounting of these units will be left to the discretion of the user. Thixotropic compound or other mechanical clamping may be used."

PAGE 23

Delete 10.1 in its entirety and substitute the following:

"10.1 Scope. This appendix details the qualification submittal plans for capacitors to be subjected to the qualification inspection specified in this specification. These plans specify the sample size, sampling criteria, and limits of coverge for the type of qualification sought."

* 10.2 Qualification categories. Delete in its entirety. (This paragraph was added by Amendment 1.)

PAGE 24

* TABLE VIII, delete and substitute:

TABLE VIII. Combined-style submittal data.

"

Style	Part number	Sample size	Rated voltage	Voltage group
CRL01 and CRL02	M83500/01-0001	9	6	I
	M83500/01-1001	9	6	
	M83500/01-0015	9	30	
	M83500/01-1015	9	30	
	M83500/01-0021	9	50	II
	M83500/01-1021	9	50	
	M83500/01-0033	9	125	
	M83500/01-1033	9	125	

"

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